

Computer Programming

Project Report



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1. Introduction

This report presents the Airline Ticket Booking System, a C++ program designed to streamline the process of booking domestic and international flights. The project showcases the application of fundamental programming concepts and problemsolving skills to build a functional and user-friendly system.

2. Objectives

To develop a program that allows users to book flights efficiently.

To calculate ticket fares dynamically based on seat type, flight type, and region.

To enhance user experience by providing meal options and discount functionality.

3. Project Design

The program design incorporates:

Input Validation: Ensures valid entries for flight type, seat type, and meal preference.

Fare Calculation: Includes base fare, taxes, discounts, and seat selection charges.

```
double calculateFare() {  
    fare = baseFare + regionFare;  
    fare *= numSeats;  
    fare *= (1 - platinumDc);  
    fare *= (1 + tax);  
  
    return fare;  
}
```

File Handling: Stores booking details for reference.

```
outFile << "Row " << setw(2) << (i / maxCols + 1) << ": ";  
for (int j = 0; j < maxCols; ++j) {  
    cout << seats[i + j] << " ";  
    outFile << seats[i + j] << " ";  
}  
cout << endl;  
outFile << endl;  
}  
  
// Close the file  
outFile.close();  
cout << "\nSeating plan has been saved to 'seating_plan.txt' successfully!" << endl;
```

Seating Plan: Provides a visual layout for seat selection.

```
Seating Plan (0: Available, X: Booked)
=====
Row 1: 0 0 0 0 0 0 0 0 0
Row 2: 0 0 0 0 0 0 0 0 0
Row 3: 0 0 0 0 0 0 0 0 0
Row 4: 0 0 0 0 0 0 0 0 0
Row 5: 0 0 0 0 0 0 0 0 0
Row 6: 0 0 0 0 0 0 0 0 0
Row 7: 0 0 0 0 0 0 0 0 0
Row 8: 0 0 0 0 0 0 0 0 0
Row 9: 0 0 0 0 0 0 0 0 0
```

Visibility and readability: Functions

```
> void displaydomestic() { ... }
> void displayinternational() { ... }
> string departureOrArrivalDate() { ... }
> void seatTypeAge(int num) { ... }
> string seatTypeClass() { ... }
> double calculateFare() { ... }
> double platinumCardDc() { ... }
> void displaySeatingPlan() { ... }
> string mealPlan() { ... }
> void displaySummary() { ... }
> void writeSummaryToFile() { ... }
```

Key Components:

Constants for tax rates, base fares, and discount values.

Variables for user inputs such as departure/arrival cities, dates, seat types, and meal preferences.

Logical structures (if/else and loops) for decision-making.

4. Implementation

Programming Language: C++ - Tools Used: Visual Studio - Core

Functionalities:

- User inputs for flight details, seat types, and meal preferences.
- Fare calculations based on multiple conditions.
- Booking summary generation.
- File handling for saving booking details.

Code Highlights:

- Input handling with loops to validate user entries.
- Fare calculation that adjusts dynamically based on seat and flight types. -
Storage of booking details in a file (booking.txt).

```
Booking Details:  
Email: momina.ali@gmail.com  
Flight Type: One-way  
Destination: mun ----to----- isb  
Departure Date: 21/12/2024  
Number of Seats: 2  
Seat Type: First Class  
Meal Preference: Default  
Total Fare: Rs 184000.00
```

5. Challenges Faced

Validating multiple user inputs while maintaining code readability.

Ensuring accurate fare calculations with varying conditions.

Designing a seating plan layout that accommodates all seats visually.

6. Lessons Learned

The importance of modular code design for easier debugging and maintenance.

Effective use of file handling for data storage.

7. Results and Output

Sample Booking Output (from booking.txt):

Email: momina.ali248@gmail.com

Flight Type: One-way

Destination: ISB ----to----- AYT

Departure Date: 24/12/2024

Number of Seats: 2

Seat Type: Business

Meal Preference: Default

Total Fare: Rs 154732.50

8. Contribution:

To ensure efficient collaboration and parallel development, we divided the implementation tasks. Aroona handled the core functionality, focusing on user inputs, such as selecting the flight type, entering city codes, and choosing preferences. This included implementing functions like `displaydomestic()` and `seatTypeAge()`. Meanwhile, Momina concentrated on fare calculation, seat booking management, and generating the booking summary. Key contributions included the functions `calculateFare()` and `writeSummaryToFile()`. This division allowed us to simultaneously work on independent parts of the program, ensuring timely completion while maintaining code quality.

9. Conclusion

The Airline Ticket Booking System demonstrates the successful application of C++ programming concepts to solve a real-world problem. By incorporating userfriendly features such as seat selection and dynamic fare calculation, the project provides a robust and efficient booking experience.

10. Future Enhancements

Integration of real-time flight availability.

Adding payment gateway functionality.

Enhancing the UI with graphical elements. (QT)